

Prime neighbours of zeta denominators

Zeta doors: a computational note

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Abstract

For positive indices $k \equiv 2 \pmod{12}$, let D_k be the reduced positive denominator of $\zeta(1-k)$. Classical Bernoulli arithmetic gives $D_k = 12T_k$, where every prime factor of T_k is congruent to 11 modulo 12. We select the unique neighbour of $D_k/3$ that is not divisible by 3 and investigate when it is prime. The construction provides known factorisations on one side of each candidate and therefore supports classical primality certificates. We give the elementary selection rule, describe its limited Lean formalisation, and record computational examples and the verification conditions needed for their interpretation. No claim of an infinite prime family or of priority for the construction is made.

1 The observation and the construction

The equality

$$\zeta(-1) = \zeta(-13) = -\frac{1}{12}$$

motivated an examination of arguments $1-k$ with $k \equiv 2 \pmod{12}$. The equality itself does not establish a period of the zeta function. Our object is the denominator at these arguments, rather than equality of their values.

For a rational number x , write $\text{den}(x)$ for its positive denominator in lowest terms. For a prime p , let $v_p(k)$ denote the exponent of p in the positive integer k . Throughout the note,

$$k \geq 2, \quad k \equiv 2 \pmod{12}, \quad D_k = \text{den}(\zeta(1-k)).$$

For an integer n not divisible by 3, put

$$\chi_3(n) = \begin{cases} 1 & n \equiv 1 \pmod{3}, \\ -1 & n \equiv 2 \pmod{3}. \end{cases}$$

For an odd prime $p \neq 3$, its *3-free door* is $m_0(p) = p + \chi_3(p)$: the even neighbour not divisible by 3.

2 Classical denominator arithmetic

With the convention $B_1 = -1/2$, the Bernoulli numbers satisfy $\zeta(1-k) = -B_k/k$ for positive even k . The classical denominator formula for these divided Bernoulli numbers is

$$D_k = \prod_{\substack{p \text{ prime} \\ p-1|k}} p^{1+v_p(k)}. \tag{1}$$

For completeness, von Staudt–Clausen gives $v_p(B_k) = -1$ when $p - 1 \mid k$. Hence $v_p(B_k/k) = -1 - v_p(k)$ for those primes. For the other primes, the classical integrality property of divided Bernoulli numbers gives $B_k/k \in \mathbb{Z}_p$; see Kellner’s account [1]. Together these facts give (1). In particular, dividing the denominator of B_k by k without accounting for cancellation would not justify this formula.

Proposition 1 (Selection rule). *If $p > 3$ is prime and $p - 1 \mid k$, then*

$$p \equiv 11 \pmod{12}, \quad m_0(p) = p - 1 \mid k.$$

Proof. The index k is even, is not divisible by 4, and is not divisible by 3. Thus neither 4 nor 3 divides $p - 1$. As p is odd, the first restriction forces $p \equiv 3 \pmod{4}$. Primality and $p > 3$ exclude $p \equiv 0 \pmod{3}$, and the second restriction excludes $p \equiv 1 \pmod{3}$. Consequently $p \equiv 2 \pmod{3}$. Combining these congruences gives $p \equiv 11 \pmod{12}$. The definition of m_0 then gives $m_0(p) = p - 1$. $\square$

Corollary 2. *There is an odd positive integer T_k such that*

$$D_k = 12T_k, \quad T_k = \prod_{\substack{p > 3 \text{ prime} \\ p-1 \mid k}} p^{1+v_p(k)}.$$

Every prime factor of T_k is 11 modulo 12. In particular, $d_k = D_k/3 = 4T_k$ is even and not divisible by 3.

Proof. In (1), 2 occurs to exponent 2, since $v_2(k) = 1$, and 3 occurs to exponent 1, since $v_3(k) = 0$. Apply Proposition 1 to the remaining factors. $\square$

Definition 3. *The zeta-door candidate at k is*

$$Q_k = d_k + \chi_3(d_k).$$

We call it a zeta-door prime when Q_k is prime.

This selects exactly one of $d_k - 1$ and $d_k + 1$: the other is divisible by 3. If Q_k is prime, its residue modulo 3 is opposite to that of d_k , so

$$m_0(Q_k) = d_k. \tag{2}$$

Writing $\Omega(T)$ for the number of prime factors counted with multiplicity, including $\Omega(1) = 0$, an equivalent expression is

$$Q_k = 4T_k + (-1)^{\Omega(T_k)}.$$

The exponent multiplicities in this expression are essential.

3 Small examples and the relation to doors

Direct rational evaluation gives the following examples.

k	$1 - k$	D_k	d_k	Q_k
2	−1	12	4	5
14	−13	12	4	5
26	−25	12	4	5
38	−37	12	4	5
50	−49	132	44	43

Thus different indices can produce the same candidate. Counts of indices, distinct denominators, and distinct primes must be separated. The table concerns reduced denominators; it does not assert that the zeta values in its first four rows are equal.

Two divisibility relations also require separation. The denominator selection rule reads $m_0(p) \mid k$ for a contributing prime $p > 3$. The ownership relation in the hire construction reads $p \mid m_0(Q)$. When $Q = Q_k$ is prime, (2) implies that every prime factor of T_k is an owner-door factor of Q . These observations do not identify the index k with a graph vertex or establish a theorem about finite gold components or Laplacian spectra.

4 Search and certification

The search starts with a fully factored index k . It enumerates all positive divisors $d \mid k$ and examines $p = d + 1$ for primality. Equation (1) then constructs D_k without computing the enormous numerator of B_k/k . Small-factor sieving and probable-prime tests screen the resulting Q_k . Screening is distinct from proof.

The targeted runs recorded here use the plus branch $Q_k = d_k + 1$. Consequently $Q_k - 1 = d_k$ has a supplied factorisation. If $N = Q_k$ and $N - 1 = FR$, where F is completely factored into proved primes and $F^2 > N$, a partial Pocklington certificate suffices. For each prime $r \mid F$, the certificate supplies a witness a_r with

$$a_r^{N-1} \equiv 1 \pmod{N}, \quad \gcd(a_r^{(N-1)/r} - 1, N) = 1.$$

These conditions force every prime divisor of N to be 1 modulo F ; since $F > \sqrt{N}$, N must be prime. This standard criterion is the primality engine, not a new primality theorem; see [2]. Minus-branch candidates would instead expose a factorisation of $Q_k + 1$ and need a suitable different certification method.

Two independent obligations remain:

1. Prove the candidate Q_k prime. Only enough proved factors of $Q_k - 1$ to meet Pocklington's bound are needed.
2. Prove the claimed denominator is exact. Every included $d + 1$ must be prime, and every omitted $d + 1$ must be composite.

For a large contributing prime p , the factorisation of $p - 1 \mid k$ is already available from that of k . A complete $p - 1$ certificate can therefore establish its primality at relatively low cost. The verifier uses a deterministic Miller–Rabin test in the range $n < 2^{64}$ and Pocklington above that range. Its seven Miller–Rabin bases are

$$2, 325, 9375, 28178, 450775, 9780504, 1795265022.$$

The range restriction is part of the verification contract.

5 Computational snapshot and reproducibility

The notebook reports the following successful candidate-primality checks in the `kitchen8` run. These are experimental records of this search, not claims of global size records.

Digits of Q_k	Distinct primes below 2^{64} in F	Larger listed factors
5522	510	5
5118	462	3
4443	360	7

For the 5522-digit entry the index is

$$k = 111695474196999960960050.$$

The author reports that all 5, 3, and 7 larger listed factors, respectively, were subsequently proved prime from their index factorisations. A separate 18365-digit candidate at $k = 1522610689958516981964050$ remains a probable prime in this snapshot; its certificate computation is not included among completed results.

The source snapshot for this draft is repository revision `8b99e2b` [3]. Change into the directory

`analysis/zeta/zeta-targeted`

and run the main reproducibility command:

```
python3 verify.py kitchen8/hits.jsonl
```

Use ordinary Python execution, without optimisation flags that disable assertions. The JSON lines contain the exact candidates, indices, factorisations, and candidate witnesses; they are preferable to printing thousands of decimal digits in the note. Reported long certificate runs have not been independently repeated during preparation of this draft.

Outstanding audit item for this draft. Revision `8b99e2b` searches bases $2, \dots, 501$ when proving large divisor-derived candidates. Exhausting this bounded search returns `False`, which the denominator loop treats as a reason to omit the candidate. Failure to find a primality witness is not a compositeness proof. A publication-ready exact-origin receipt must distinguish *prime*, *composite*, and *unresolved*, and must not finish successfully with unresolved divisor candidates. The successful proofs of all included large factors do not by themselves settle this exhaustiveness issue. This qualification concerns denominator provenance, not the separate partial-Pocklington proof of Q_k .

6 Formalisation and further questions

The file `lean/Hire/ZetaDoors.lean` contains the selection theorem

`prime_mod_twelve_of_pred_dvd_index`

and its door-divisibility consequence `m0_dvd_of_pred_dvd_index`. The first proves Proposition 1’s residue conclusion from primality, $p \neq 2, 3$, $k \bmod 12 = 2$, and $p - 1 \mid k$; the second rewrites the door as $p - 1$. The file mentions Mathlib’s von Staudt–Clausen theorem but does not invoke it. It does not formalise (1), the complete search, or the large numerical primality certificates.

The immediate computational questions are how often denominators repeat, how the two signs are distributed, and how prime frequency changes under different index-selection schemes. A density measured in a targeted sample of highly factored indices should not be presented as the density over all $k \equiv 2 \pmod{12}$. Whether infinitely many distinct zeta-door primes exist is not settled here. A literature comparison is also needed before making a priority claim for this particular construction.

The companion Lab at <https://halfasecond.com/> retains changing records and experiments; this note fixes definitions, proof scope, and a dated computational snapshot.

References

- [1] B. C. Kellner, *The Bernoulli Number Page*, sections on denominator and numerator structure, <https://www.bernoulli.org/>. Accessed 22 September 2026.
- [2] PrimePages, *Finding primes and proving primality*, Section 3: classical primality tests, <https://t5k.org/prove/prove3.html>.
- [3] J. Pickett, *Natural Maths notebook*, source and computational records, revision 8b99e2b, <https://github.com/hasjack/nm-notebook/tree/8b99e2b>.